Sunspot Number V2.0 Through Solar Cycle 25: A Long-term Multi-proxy Stability Analysis

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Guiding question

Does SN V2.0 show coherent deviations relative to independent solar-activity proxies?

The Sunspot Number is the longest direct index of solar activity, but it is a network-constructed count-based record. Its long-term stability can therefore be affected by observer practice, coverage, instrumentation, and network evolution.

[Context: Clette & Lefèvre (2016); Clette et al. (2023); SILSO Sunspot Number V2.0.]

Here we test SN V2.0 against independent solar-activity proxies that sample different physical layers: sunspot area, F10.7 radio flux, Mg II, LOS magnetic field, and AAVSO Ra.

Poster logic: Detect coherent SN–proxy departures and test whether the strongest modern departure is better explained by solar structure or index/proxy calibration effects.

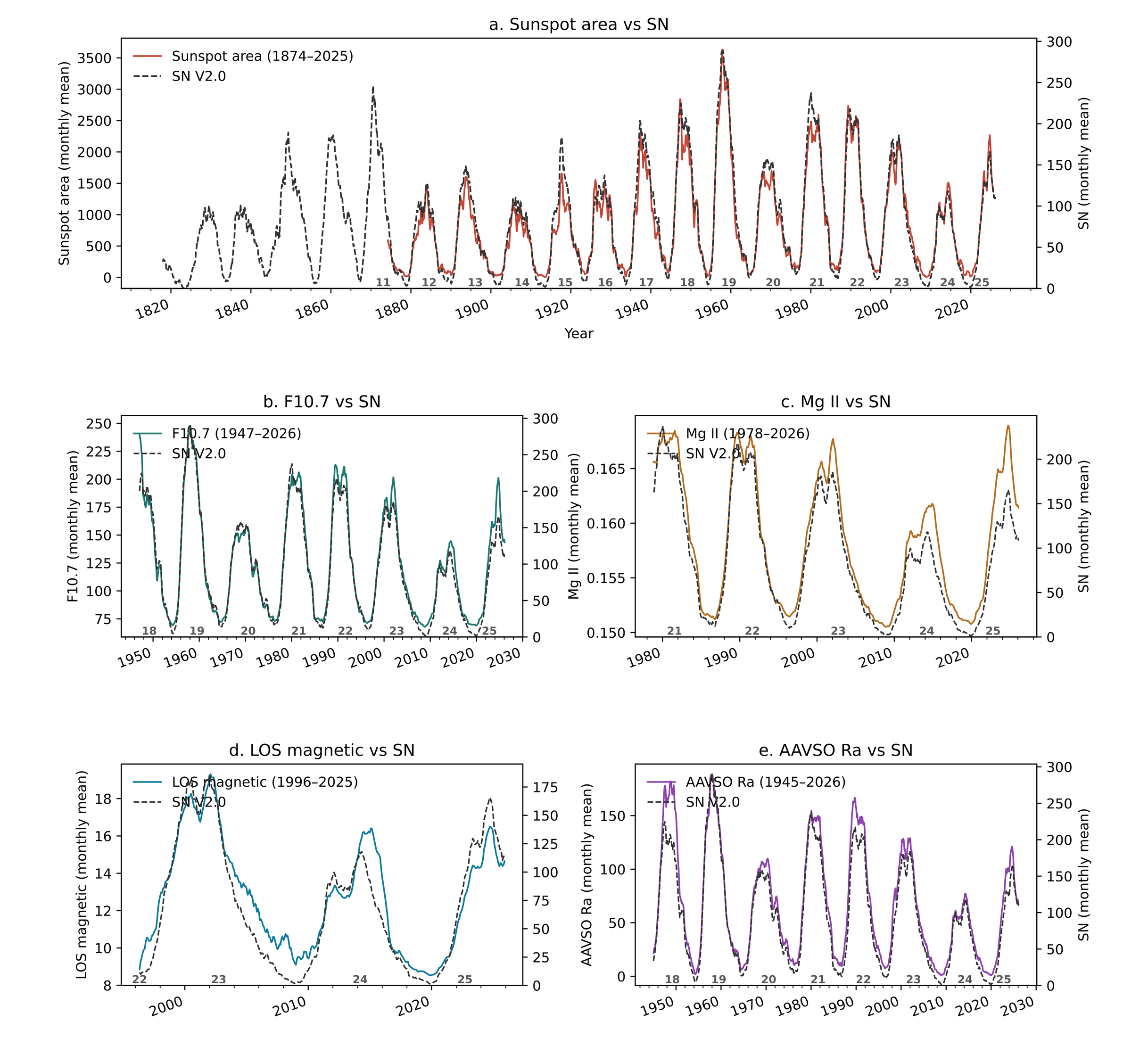
Independent proxies used

Proxy	Source	Span	Layer / role
SN V2.0	SILSO	1818–present	Reference
Sunspot area	Mandal	1874–2025	photospheric geometry
F10.7	Penticton / NOAA	1947–2026	radio upper atmosphere
Mg II	Bremen / LASP	1978–2026	UV / chromosphere
LOS magnetic field	MDI / HMI	1996–2025	disk magnetic field
AAVSO Ra	AAVSO	1945–2026	independent counts

Method pipeline

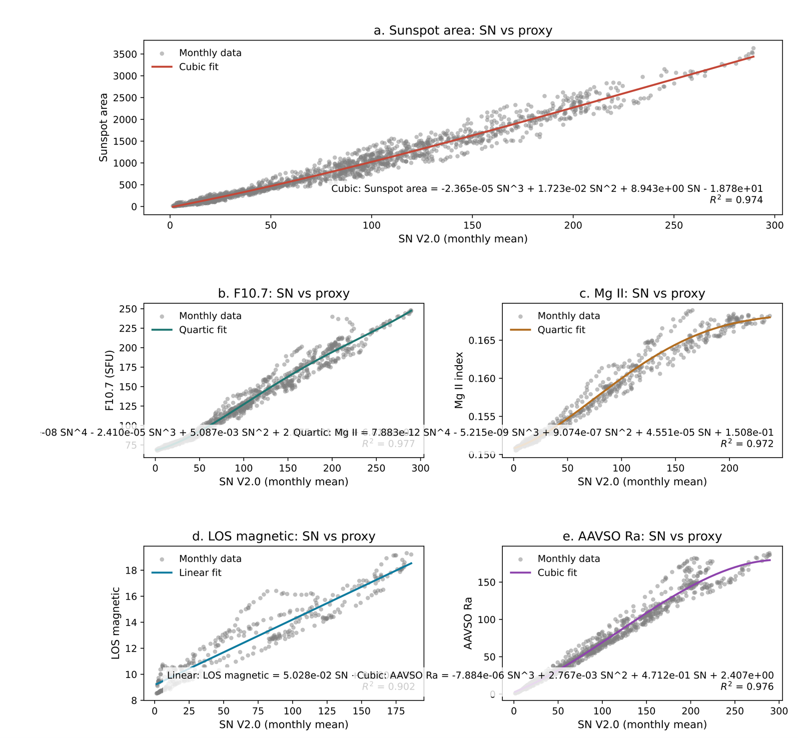
1. Daily records → monthly means
2. Centered 10-month running mean
3. Proxy-specific SN→proxy fit
4. Reconstruct expected proxy
5. Compute fractional residual $\beta(t)$
6. Median/MAD standardization
7. Persistent and coherent detection of deviations

Monthly backbone



All proxies follow the solar cycle during their overlap intervals.

Why nonlinear fits required



Proxy response to SN is not always linear.

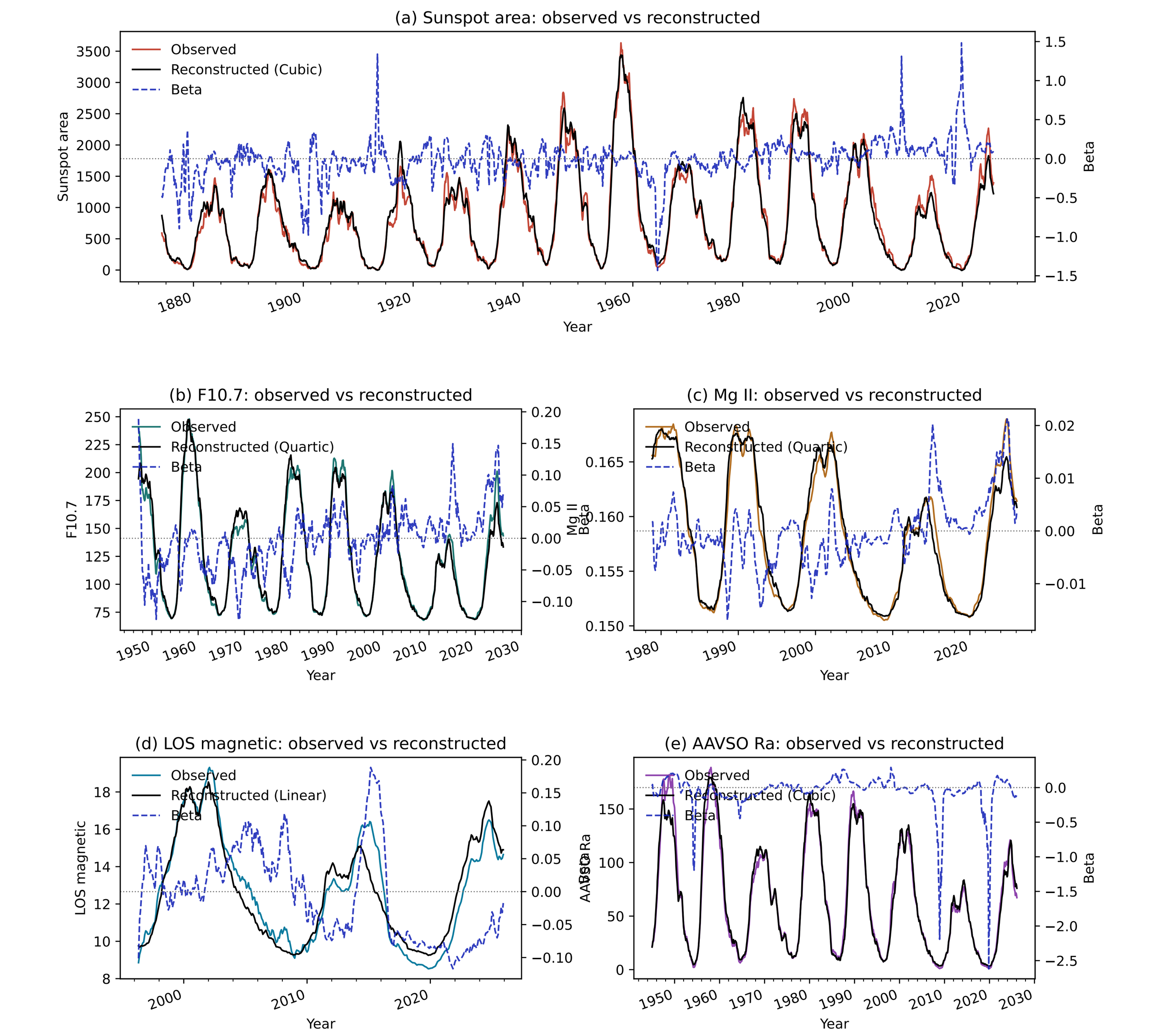
Proxy	Fit
Area	Cubic
F10.7	Quartic
Mg II	Quartic
LOS magnetic	Linear
AAVSO R_a	Cubic

The fit is used to remove the expected proxy response before searching for departures.

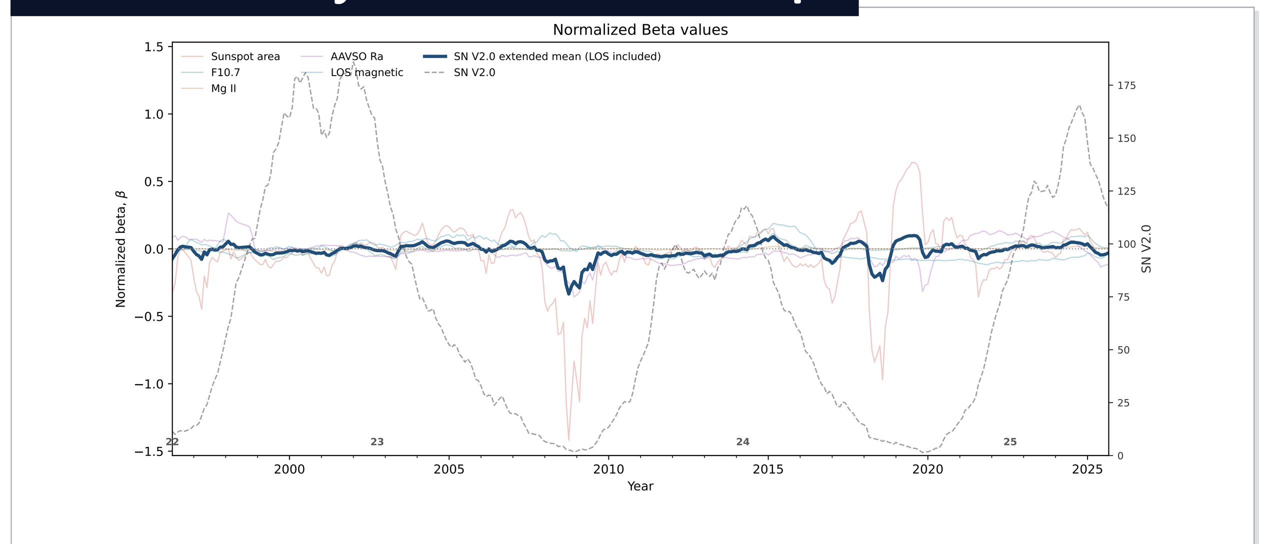
Beta diagnostic

$$\beta_j(t) = \frac{X_j(t) - \hat{X}_j(t)}{\hat{X}_j(t)}$$

- $X_j(t)$ is observed; $\hat{X}_j(t)$ is predicted from SN.
- $\beta \approx 0$ indicates expected SN–proxy behavior.



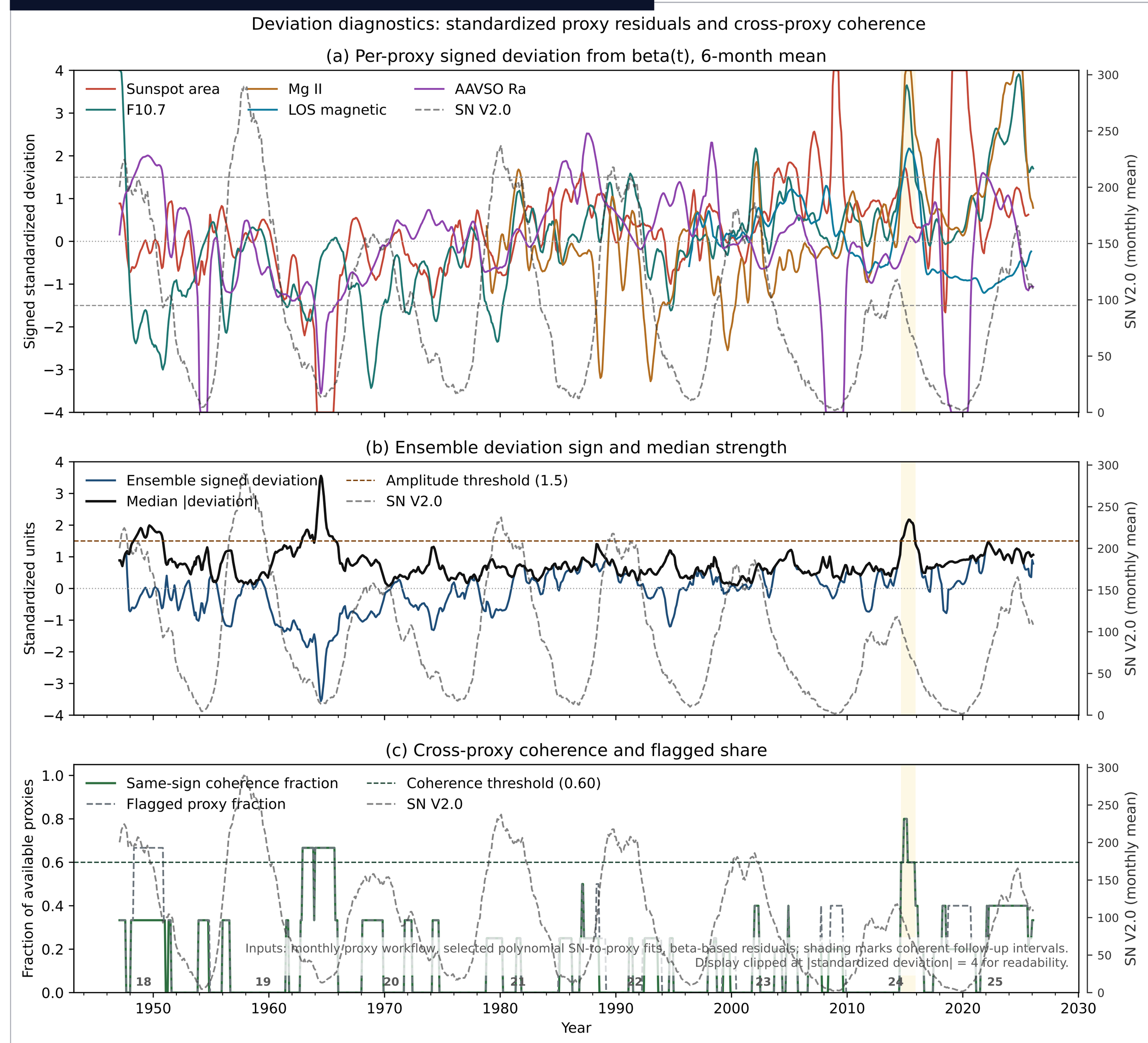
Broad stability on common overlaps



1978–2025	−0.026	No collapse
Overlap interval	Mean β	Broadly stable SN

The common-overlap beta ensemble remains near zero, while individual proxies show cycle and layer-dependent deviations.

Coherent departure detection



Detection rule:

- Amplitude ≥ 1.5, Coherence ≥ 0.60;
- at least 3 coherent proxies for at least 4 months.

Summary of deviation detection

Does SN V2.0 show coherent deviations relative to independent solar-activity proxies?

- Yes. But, deviations are confined to one prominent coherent interval.
- Interval: 2014-09 to 2015-11, declining phase of solar cycle 24.
- Proxies: F10.7, LOS magnetic, Mg II, and sunspot area.

Next Question: Are these deviations solar-origin?

Literature context

1. SN–F10.7 / Mg II

- **Calibration view:** Clette argues for an F10.7 inhomogeneity near 1980–1981 [Clette2021].
- **Solar-physics view:** Mursula et al. argue that changing SN–F10.7 relations reflect real changes in the solar atmosphere, not simply F10.7 calibration [Mursula2024].
- Okoh and Okoro identify 2014–2015 as a conspicuous high-F10.7 interval relative to SN [Okoh2020].
- Svalgaard frames the 2014–2015 SN–F10.7 anomaly as a possible SN-derivation problem [Svalgaard2025].
- Radio, UV, and SN relations are known to depend on cadence, averaging, cycle phase, and atmospheric layer [Tapping2013; Bruevich2014; Floyd2005].

2. SN–area / morphology

- Tapping and Morgan show that the relation between SN, total sunspot area, and F10.7 changed noticeably in Cycles 21–24 [TappingMorgan2017].
- A changing sunspot-size distribution can naturally change the SN–area mapping.
- Deficits of small spots and multiple sunspot populations support a cycle-dependent SN–area relation [LefevreClette2011; Nagovitsyn2016] [MunozJaramillo2015; NagovitsynPevtsov2021].
- Long-term changes in spots per group question the assumption that count-based and group-based indices preserve a fixed relation [Pevtsov2025].

The literature allows both interpretations: Recent proxy and morphology studies increasingly support a solar-physics contribution to the 2014–2015 deviation.

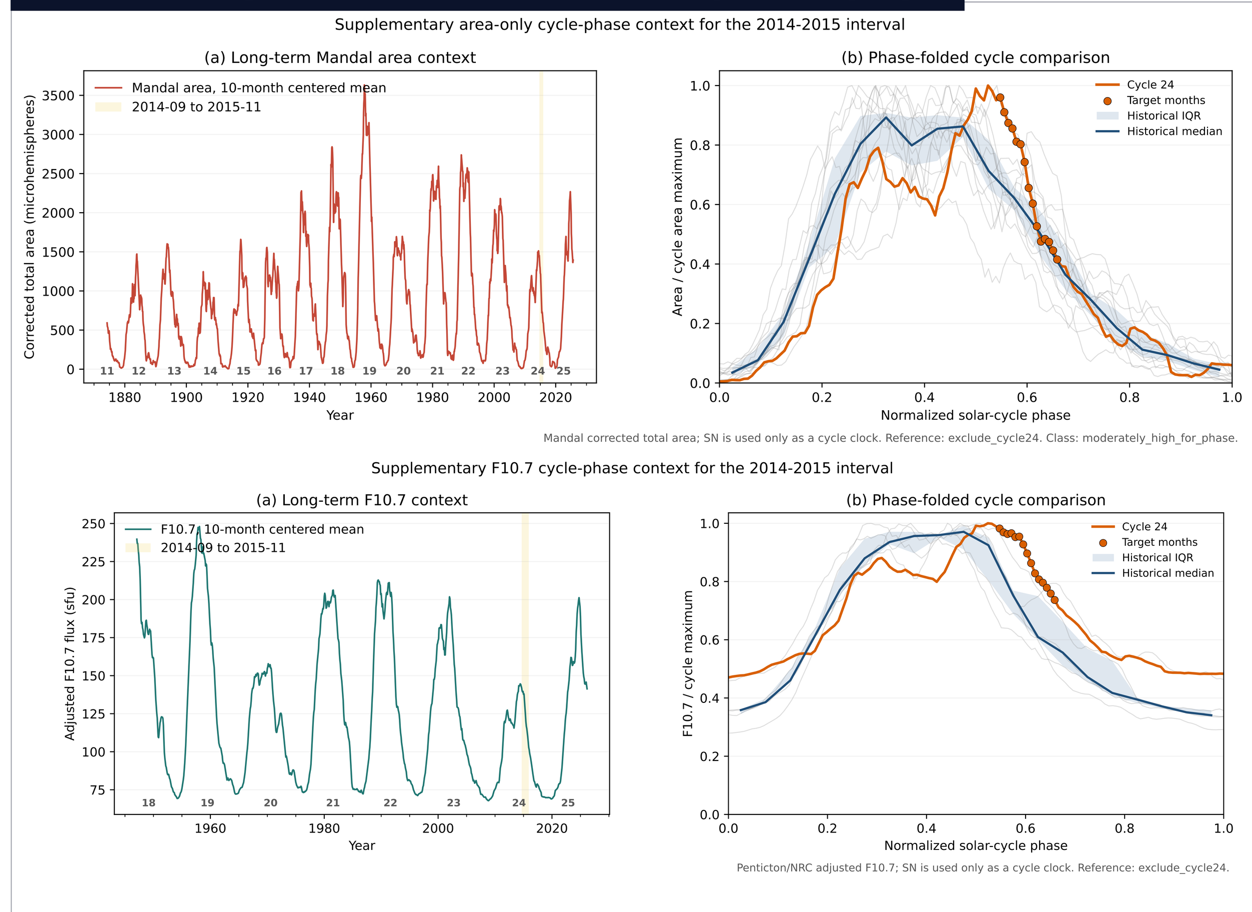
Follow-up

Q1 Are the 2014–2015 Proxies high relative to comparable phases of other solar cycles of the proxy?

Q2 Are high proxy/SN values accompanied by real photospheric or magnetic structures?

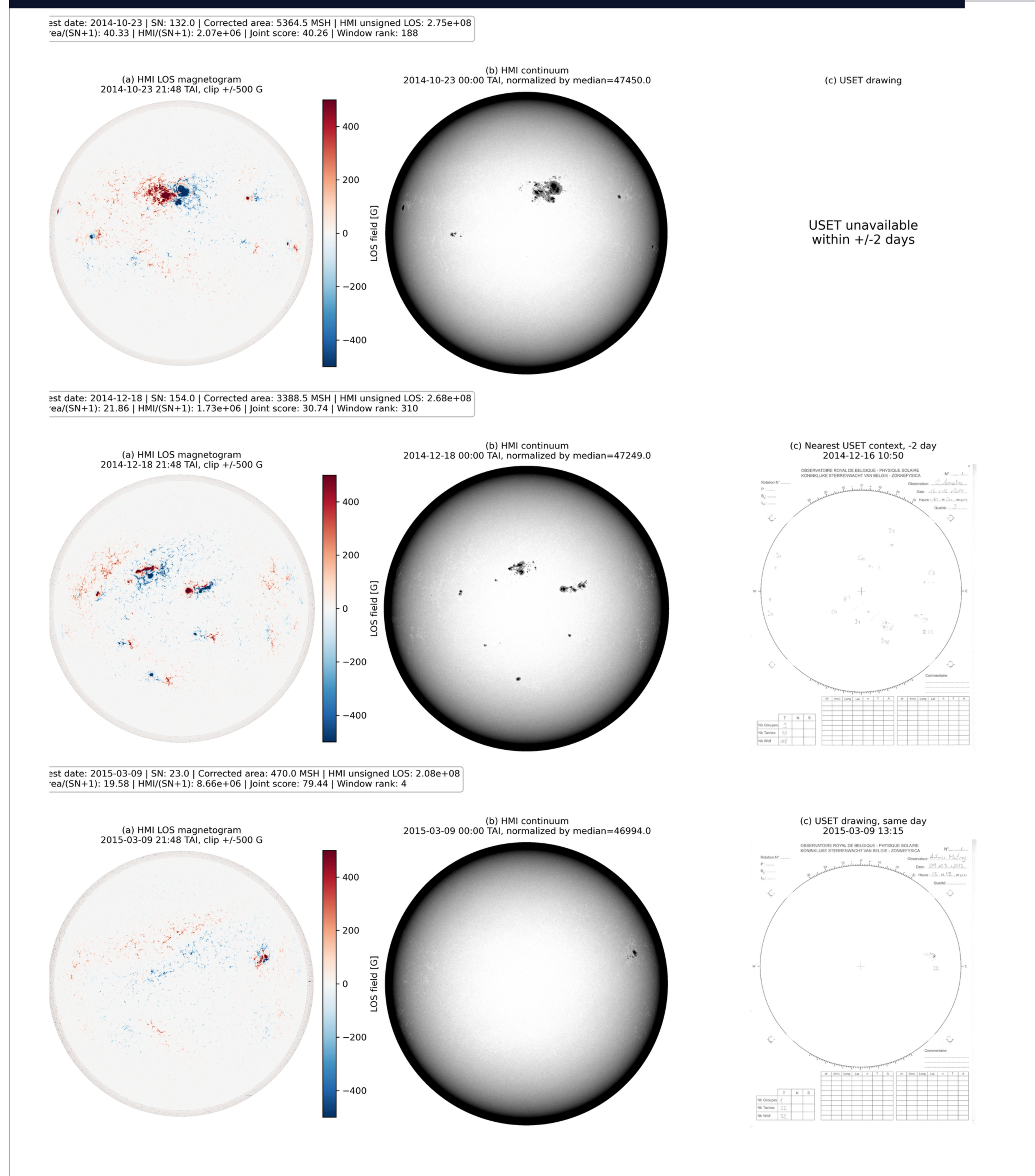
Use independent images and regions to assess whether the deviations are plausibly of solar origin.

Was the Sun more active in 2014–2015?



During 2014–2015, cycle 24 shows persistently higher area and F10.7 levels than comparable phases of earlier solar cycles.

Larger and Magnetically Stronger Active Regions at Comparable SN



HMI LOS magnetograms, continuum images, and USET drawings show that the 2014–2015 proxy excess corresponds to real photospheric and magnetic structures—large sunspots and strong active-region fields.

Conclusions

- ✓ SN V2.0 is broadly stable across independent solar-activity proxies over long, overlapping periods.
- ✓ A clear modern departure occurs mainly in 2014–2015 of Solar Cycle 24 across several proxies.
- ✓ The combined proxy, cycle-phase, magnetic, and image evidence favors a solar-origin explanation tied to unusually large, strongly magnetized active regions.